

Fig. 2: Create Project window

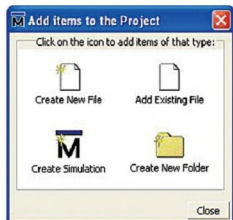


Fig. 3: Add Items to the Project window

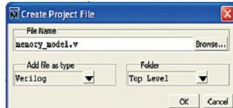


Fig. 4: Create Project File window

a functional test, a stuck-type fault model in logic circuits and their pattern generation for read-and-write operations will be demonstrated by writing a test bench for the memory model and BIST controller on ModelSim.

In this project we will test BIST for stuck-at-fault 0 for the memory model. The design hierarchy of BIST design is as follows:

- tb.v
- bist_controller.v
- memory_model.v

Here, test bench (tb.v) will generate a pattern for all memory locations with the help of a pattern generator defined in bist_controller.v,

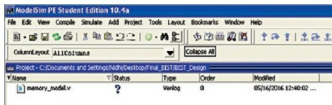


Fig. 5: Workspace window

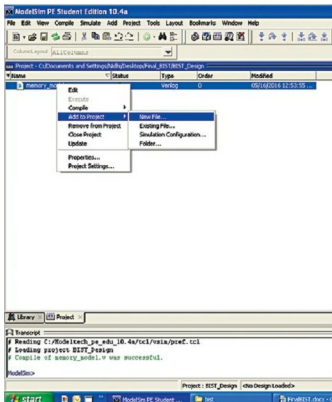


Fig. 6: Adding new files

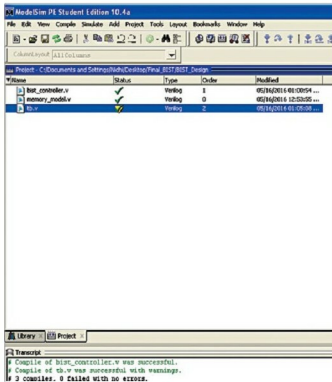


Fig. 7: Workspace station

which will further generate the pass or fail status as per the finite-state machine.

You need to provide inputs manually to Memory_model.v when not in BIST mode. So, for testing of BIST for stuck-at-fault 0, we need to simulate the above-mentioned three files. Additionally, we need sim.do simulation file. It should be present in the same folder of the above-mentioned files.

Software program

Verilog programming is used in this project. Bottom-up design has been followed in this article in order to facilitate easy debugging of various modules.

Modelsim is an easy-to-use yet versatile VHDL/System-Verilog/Verilog/SystemC simulator from Mentor Graphics. It supports behavioural, register-transfer-level and gate-level modelling.

To start BIST design and memory model simulation, install ModelSim V10.4a on a Windows PC and follow the steps mentioned below:

Create the project.

1. Start ModelSim from the desktop; you will see ModelSim 10.4a dialogue window.

2. Create a project by clicking Jump